Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Azevedo Dairy Inc

Physical address of dairy:

1337 Magnolia AVE	Patterson	Stanislaus	95363
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1916

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X047-X007-X006-XXXX

B. OPERATORS

Azevedo, John M

Operator name: Azevedo, John M

Telephone no.: (209) 892-6961 Landline (209) 495-3054 Cellular

1337 Magnolia AVE	Patterson	CA	95636
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Azevedo, John M

Legal owner name: Azevedo, John M

Telephone no.: (209) 892-6961 Landline (209) 495-3054 Cellular

1337 Magnolia AVE	Patterson	CA	95636
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	48	38	0	68	0
Number under roof	460	0	0	0	0	0
Maximum number	460	48	38	0	68	0
Average number	460	48	38	0	68	0
Avg live weight (lbs)	1,450	1,500	1,000	0		

Predominant milk cow breed: Holstein

Average milk production: 79 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 13,632.50 *tons per reporting period*

Total nitrogen from manure: 176,468.51 *lbs per reporting period*

After ammonia losses (30% loss applied): 123,527.96 *lbs per reporting period*

Total phosphorus from manure: 29,298.20 *lbs per reporting period*

Total potassium from manure: 90,814.22 *lbs per reporting period*

Total salt from manure: 227,628.60 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,071,250 *gallons*

Total nitrogen generated: 12,302.20 *lbs*

Total phosphorus generated: 1,550.59 *lbs*

Total potassium generated: 16,428.56 *lbs*

Total salt generated: 133,786.41 *lbs*

	3,071,250	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	3,071,250	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - 2014 Barn Well	Ground water
Domestic Well - Calf/House Barn	Ground water
Irrigation Water - Canal	Surface water
Irrigation Water - Dairy Deep Well	Ground water
Irrigation Water - Lemon Well	Ground water

Source Description	Type
Irrigation Water - Magnolia Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	Corral solids	1,800.00 *ton*	Dry-weight	24.5		15,900.00	5,100.00	15,600.00		0.00
10/08/2023	Corral solids	2,000.00 *ton*	Dry-weight	32.6		17,700.00	6,100.00	29,700.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	90,935.40	30,307.40	122,472.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	90,935.40	30,307.40	122,472.00	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Costa Grande	49	49	0	none	X047-X027-X015-XXXX
Dairy	30	30	2	process wastewater	X047-X007-X006-XXXX
Greg's	35	35	6	manure	X047-X011-X011-XXXX
Heifer Ranch	48	48	2	manure	X047-X011-X013-XXXX
Little Costa/Schut	40	40	6	manure	X047-X010-X019-XXXX
Poolman	30	30	2	manure	X047-X010-X021-XXXX
Thorkelson	42	42	0	none	X047-X011-X012-XXXX
Totals for areas that were used for application	183	183	18		
Totals for areas that were not used for application	91	91	0		
Land application area totals	274	274	18		

B. CROPS AND HARVESTS

Dairy

Field name: Dairy

12/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 12/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	456.30 *ton*	Dry-weight		66.5	15,600.00	2,500.00	15,700.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.21	158.97	25.48	159.99	1,070.02

Dairy

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	912.50 *ton*	Dry-weight		63.1	9,600.00	2,100.00	11,900.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.42	215.50	47.14	267.13	1,301.96

Greg's

Field name: Greg's

10/22/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 35 Plant date: 10/22/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	53.20 *ton*	Dry-weight		64.1	39,700.00	2,900.00	22,100.00		9.60
06/06/2023	49.80 *ton*	Dry-weight		64.1	39,700.00	2,900.00	22,100.00		9.60
07/04/2023	48.50 *ton*	Dry-weight		64.1	39,700.00	2,900.00	22,100.00		9.60
08/02/2023	43.60 *ton*	Dry-weight		64.1	39,700.00	2,900.00	22,100.00		9.60
08/30/2023	42.30 *ton*	Dry-weight		64.1	39,700.00	2,900.00	22,100.00		9.60
10/03/2023	43.20 *ton*	Dry-weight		64.1	39,700.00	2,900.00	22,100.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.02	228.53	16.69	127.21	552.61

Heifer Ranch

Field name: Heifer Ranch

Heifer Ranch

12/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 48 Plant date: 12/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	612.30 *ton*	Dry-weight		66.1	15,800.00	2,600.00	16,000.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.76	136.65	22.49	138.38	1,003.25

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 48 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	1,480.50 *ton*	Dry-weight		62.7	9,800.00	2,000.00	10,200.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.84	225.49	46.02	234.70	1,173.48

Little Costa/Schut

Field name: Little Costa/Schut

Little Costa/Schut

05/27/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 40 Plant date: 05/27/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	80.20 *ton*	Dry-weight		61.4	40,800.00	2,900.00	22,000.00		9.10
06/06/2023	49.60 *ton*	Dry-weight		61.4	40,800.00	2,900.00	22,000.00		9.10
07/04/2023	49.50 *ton*	Dry-weight		61.4	40,800.00	2,900.00	22,000.00		9.10
08/02/2023	50.00 *ton*	Dry-weight		61.4	40,800.00	2,900.00	22,000.00		9.10
08/30/2023	48.90 *ton*	Dry-weight		61.4	40,800.00	2,900.00	22,000.00		9.10
10/03/2023	49.50 *ton*	Dry-weight		61.4	40,800.00	2,900.00	22,000.00		9.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.19	258.04	18.34	139.14	575.54

Poolman

Field name: Poolman

12/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 12/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	475.60 *ton*	Dry-weight		65.7	13,300.00	1,800.00	13,900.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.85	144.64	19.58	151.17	880.91

Poolman

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	912.30 *ton*	Dry-weight		62.8	9,600.00	2,000.00	10,500.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.41	217.20	45.25	237.56	1,267.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dairy - 12/03/2022: Oats, silage-soft dough

Field name: Dairy

Crop: Oats, silage-soft dough — Plant date: 12/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Dairy Deep Well	Ground water	44.10	0.00	0.00	2,489.17	5,523,750.00 *gal*
Application event totals		44.10	0.00	0.00	2,489.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	122.57	15.45	163.68	1,332.96	918,000.00 *gal*
Irrigation Water - Dairy Deep Well	Ground water	43.92	0.00	0.00	2,479.03	5,501,250.00 *gal*
Application event totals		166.49	15.45	163.68	3,811.99	

Dairy - 05/30/2023: Corn, silage

Field name: Dairy

Crop: Corn, silage — Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.39	0.00	0.00	118.35	4,624,700.00 *gal*
Application event totals		0.39	0.00	0.00	118.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.51	4,552,860.00 *gal*
Application event totals		0.38	0.00	0.00	116.51	

Dairy - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	117.43	4,588,780.00 *gal*
Application event totals		0.38	0.00	0.00	117.43	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.25	13.52	143.22	1,166.34	803,250.00 *gal*
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.51	4,552,860.00 *gal*
Application event totals		107.63	13.52	143.22	1,282.86	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.43	12.15	128.78	1,048.73	722,250.00 *gal*
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.51	4,552,860.00 *gal*
Application event totals		96.81	12.15	128.78	1,165.24	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.82	10.56	111.93	911.51	627,750.00 *gal*
Irrigation Water - Canal	Surface water	0.39	0.00	0.00	118.12	4,615,720.00 *gal*
Application event totals		84.20	10.56	111.93	1,029.63	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.28	4,543,880.00 *gal*
Application event totals		0.38	0.00	0.00	116.28	

Dairy - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.05	4,534,900.00 *gal*
Application event totals		0.38	0.00	0.00	116.05	

Greg's - 10/22/2020: Alfalfa, hay

Field name: Greg's

Crop: Alfalfa, hay

Plant date: 10/22/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.40	0.00	0.00	123.70	5,639,440.00 *gal*
Application event totals		0.40	0.00	0.00	123.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.40	0.00	0.00	121.93	5,558,620.00 *gal*
Application event totals		0.40	0.00	0.00	121.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.39	0.00	0.00	120.75	5,504,740.00 *gal*
Application event totals		0.39	0.00	0.00	120.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.40	0.00	0.00	123.90	5,648,420.00 *gal*
Application event totals		0.40	0.00	0.00	123.90	

Greg's - 10/22/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.40	0.00	0.00	123.51	5,630,460.00 *gal*
Application event totals		0.40	0.00	0.00	123.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.39	0.00	0.00	120.75	5,504,740.00 *gal*
Application event totals		0.39	0.00	0.00	120.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	238.60	82.23	400.36	0.00	350.00 *ton*
Application event totals		238.60	82.23	400.36	0.00	

Heifer Ranch - 12/03/2022: Oats, silage-soft dough

Field name: Heifer Ranch

Crop: Oats, silage-soft dough

Plant date: 12/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	170.56	54.71	167.35	0.00	341.00 *ton*
Application event totals		170.56	54.71	167.35	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Magnolia Well	Ground water	0.14	0.00	0.00	69.28	4,151,250.00 *gal*
Application event totals		0.14	0.00	0.00	69.28	

Heifer Ranch - 12/03/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Lemon Well	Ground water	12.89	0.00	0.00	1,201.94	4,164,750.00 *gal*
Application event totals		12.89	0.00	0.00	1,201.94	

Heifer Ranch - 05/30/2023: Corn, silage

Field name: Heifer Ranch

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	253.09	81.18	248.32	0.00	506.00 *ton*
Application event totals		253.09	81.18	248.32	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Lemon Well	Ground water	12.85	0.00	0.00	1,198.04	4,151,250.00 *gal*
Application event totals		12.85	0.00	0.00	1,198.04	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Magnolia Well	Ground water	0.15	0.00	0.00	69.73	4,178,250.00 *gal*
Application event totals		0.15	0.00	0.00	69.73	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Lemon Well	Ground water	12.87	0.00	0.00	1,199.99	4,158,000.00 *gal*
Application event totals		12.87	0.00	0.00	1,199.99	

Heifer Ranch - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Magnolia Well	Ground water	0.14	0.00	0.00	69.17	4,144,500.00 *gal*
Application event totals		0.14	0.00	0.00	69.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Lemon Well	Ground water	12.83	0.00	0.00	1,196.09	4,144,500.00 *gal*
Application event totals		12.83	0.00	0.00	1,196.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Magnolia Well	Ground water	0.14	0.00	0.00	69.17	4,144,500.00 *gal*
Application event totals		0.14	0.00	0.00	69.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Lemon Well	Ground water	12.85	0.00	0.00	1,198.04	4,151,250.00 *gal*
Application event totals		12.85	0.00	0.00	1,198.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Magnolia Well	Ground water	0.15	0.00	0.00	70.07	4,198,500.00 *gal*
Application event totals		0.15	0.00	0.00	70.07	

Little Costa/Schut - 05/27/2020: Alfalfa, hay

Field name: Little Costa/Schut

Crop: Alfalfa, hay

Plant date: 05/27/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Little Costa/Schut - 05/27/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.41	0.00	0.00	125.30	6,528,460.00 *gal*
Application event totals		0.41	0.00	0.00	125.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.41	0.00	0.00	124.96	6,510,500.00 *gal*
Application event totals		0.41	0.00	0.00	124.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.41	0.00	0.00	124.61	6,492,540.00 *gal*
Application event totals		0.41	0.00	0.00	124.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.41	0.00	0.00	125.30	6,528,460.00 *gal*
Application event totals		0.41	0.00	0.00	125.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.41	0.00	0.00	125.30	6,528,460.00 *gal*
Application event totals		0.41	0.00	0.00	125.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.41	0.00	0.00	124.96	6,510,500.00 *gal*
Application event totals		0.41	0.00	0.00	124.96	

Little Costa/Schut - 05/27/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	262.46	90.45	440.39	0.00	440.00 *ton*
Application event totals		262.46	90.45	440.39	0.00	

Poolman - 12/03/2022: Oats, silage-soft dough

Field name: Poolman

Crop: Oats, silage-soft dough

Plant date: 12/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	193.67	62.12	190.02	0.00	242.00 *ton*
Application event totals		193.67	62.12	190.02	0.00	

03/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.46	0.00	0.00	141.94	5,546,250.00 *gal*
Application event totals		0.46	0.00	0.00	141.94	

04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.45	0.00	0.00	139.06	5,433,750.00 *gal*
Application event totals		0.45	0.00	0.00	139.06	

Poolman - 05/30/2023: Corn, silage

Field name: Poolman

Crop: Corn, silage

Plant date: 05/30/2023

Poolman - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	288.70	99.50	484.43	0.00	363.00 *ton*
Application event totals		288.70	99.50	484.43	0.00	

06/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.05	4,534,900.00 *gal*
Application event totals		0.38	0.00	0.00	116.05	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.51	4,552,860.00 *gal*
Application event totals		0.38	0.00	0.00	116.51	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	117.43	4,588,780.00 *gal*
Application event totals		0.38	0.00	0.00	117.43	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.97	4,570,820.00 *gal*
Application event totals		0.38	0.00	0.00	116.97	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.97	4,570,820.00 *gal*
Application event totals		0.38	0.00	0.00	116.97	

Poolman - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	117.43	4,588,780.00 *gal*
Application event totals		0.38	0.00	0.00	117.43	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	116.51	4,552,860.00 *gal*
Application event totals		0.38	0.00	0.00	116.51	

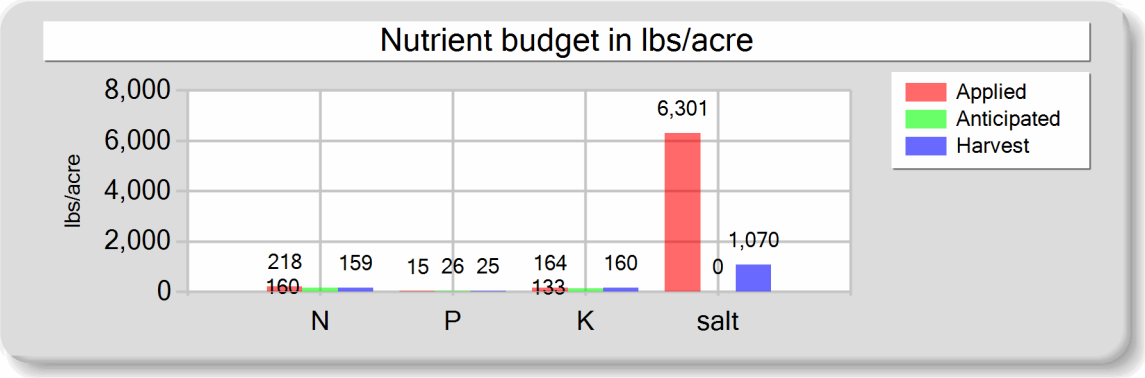
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.38	0.00	0.00	117.43	4,588,780.00 *gal*
Application event totals		0.38	0.00	0.00	117.43	

B. NUTRIENT BUDGET

Dairy - 12/03/2022: Oats, silage-soft dough

Field name: Dairy | Crop: Oats, silage-soft dough | Plant date: 12/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	122.57	15.45	163.68	1,332.96
Fresh water	88.02	0.00	0.00	4,968.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.59	15.45	163.68	6,301.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	158.97	25.48	159.99	1,070.02
Nutrient balance	58.61	-10.03	3.69	5,231.14
Applied to removed ratio	1.37	0.61	1.02	5.89

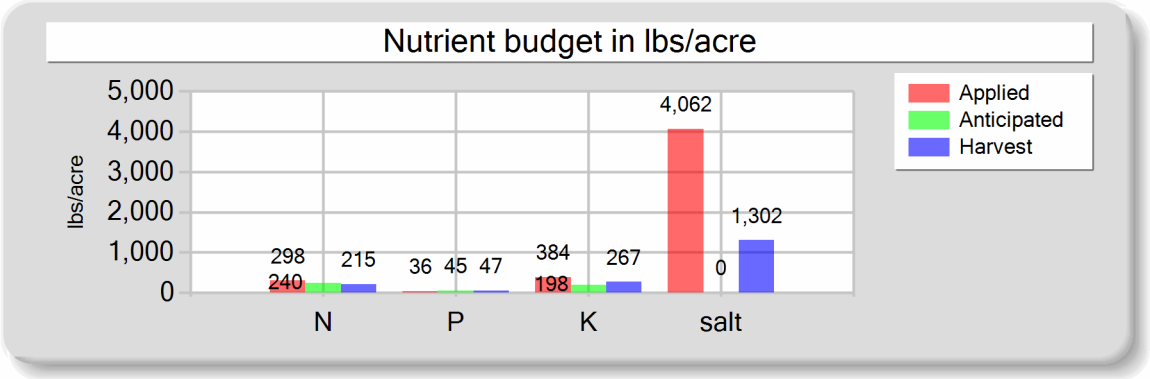
Fresh water applied	
11,025,000.00	*gallons*
406.01	*acre-inches*
13.53	*inches/acre*

Process wastewater applied	
918,000.00	*gallons*
33.81	*acre-inches*
1.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy - 05/30/2023: Corn, silage

Field name: Dairy | Crop: Corn, silage | Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	287.50	36.24	383.93	3,126.58
Fresh water	3.05	0.00	0.00	935.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.55	36.24	383.93	4,062.37
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	215.50	47.14	267.13	1,301.96
Nutrient balance	82.06	-10.90	116.81	2,760.42
Applied to removed ratio	1.38	0.77	1.44	3.12

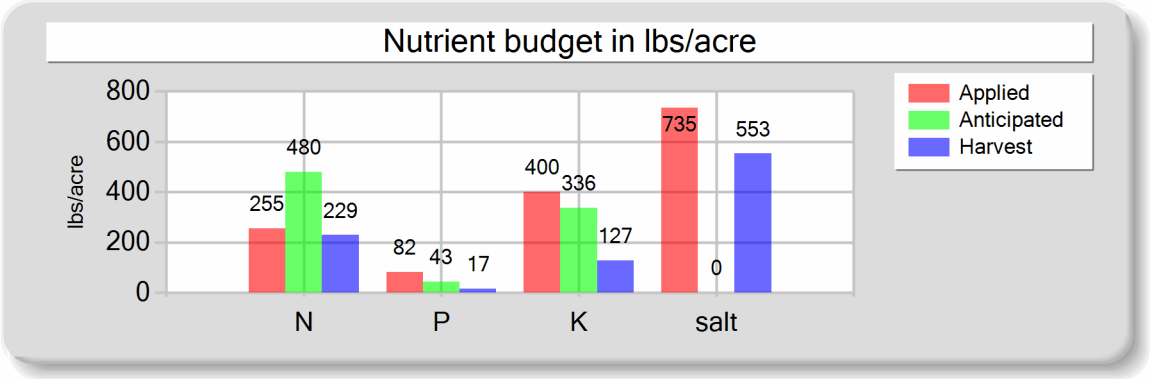
Fresh water applied	
36,566,560.00	*gallons*
1,346.62	*acre-inches*
44.89	*inches/acre*

Process wastewater applied	
2,153,250.00	*gallons*
79.30	*acre-inches*
2.64	*inches/acre*

Total harvests for the crop	
1	*harvests*

Greg's - 10/22/2020: Alfalfa, hay

Field name: Greg's Crop: Alfalfa, hay Plant date: 10/22/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	238.60	82.23	400.36	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.40	0.00	0.00	734.54
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	254.99	82.23	400.36	734.54
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	228.53	16.69	127.21	552.61
Nutrient balance	26.47	65.53	273.14	181.93
Applied to removed ratio	1.12	4.93	3.15	1.33

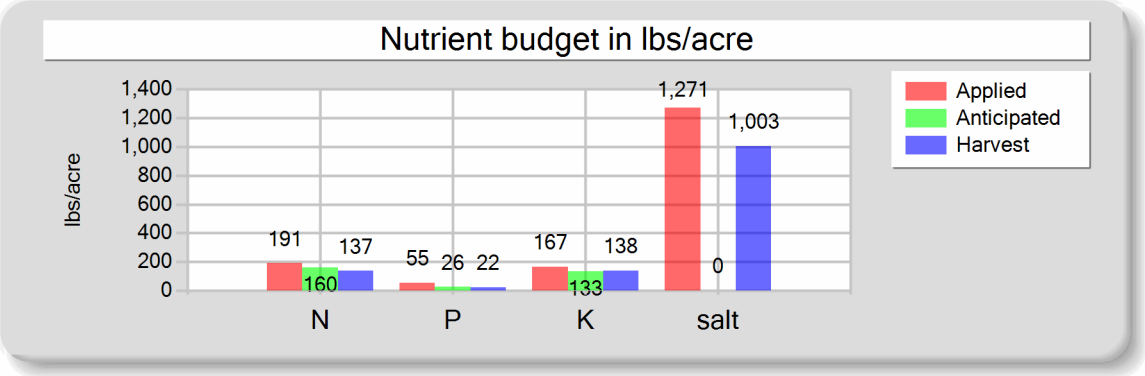
Fresh water applied	
33,486,420.00	*gallons*
1,233.19	*acre-inches*
35.23	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
6	*harvests*

Heifer Ranch - 12/03/2022: Oats, silage-soft dough

Field name: Heifer Ranch | Crop: Oats, silage-soft dough | Plant date: 12/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	170.56	54.71	167.35	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.03	0.00	0.00	1,271.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.60	54.71	167.35	1,271.22
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	136.65	22.49	138.38	1,003.25
Nutrient balance	53.95	32.22	28.97	267.97
Applied to removed ratio	1.39	2.43	1.21	1.27

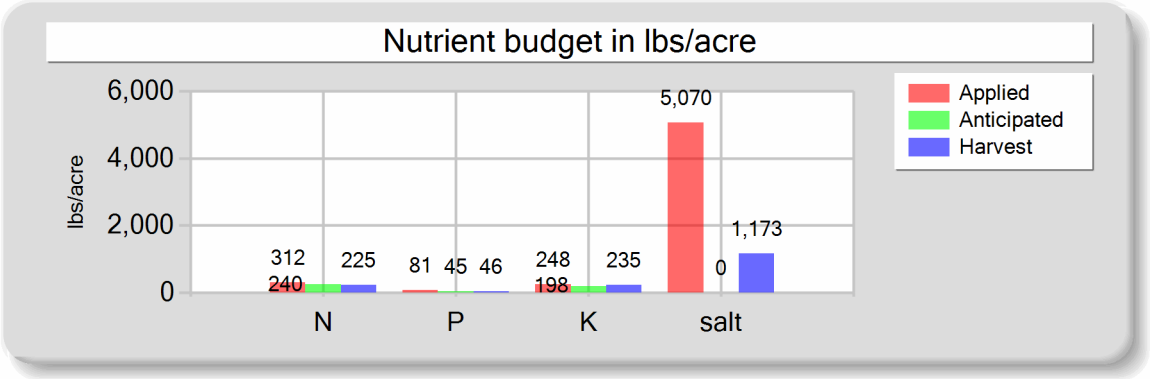
Fresh water applied	
8,316,000.00	*gallons*
306.25	*acre-inches*
6.38	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Heifer Ranch - 05/30/2023: Corn, silage

Field name: Heifer Ranch | Crop: Corn, silage | Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	253.09	81.18	248.32	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	51.97	0.00	0.00	5,070.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.06	81.18	248.32	5,070.32
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.49	46.02	234.70	1,173.48
Nutrient balance	86.57	35.16	13.62	3,896.84
Applied to removed ratio	1.38	1.76	1.06	4.32

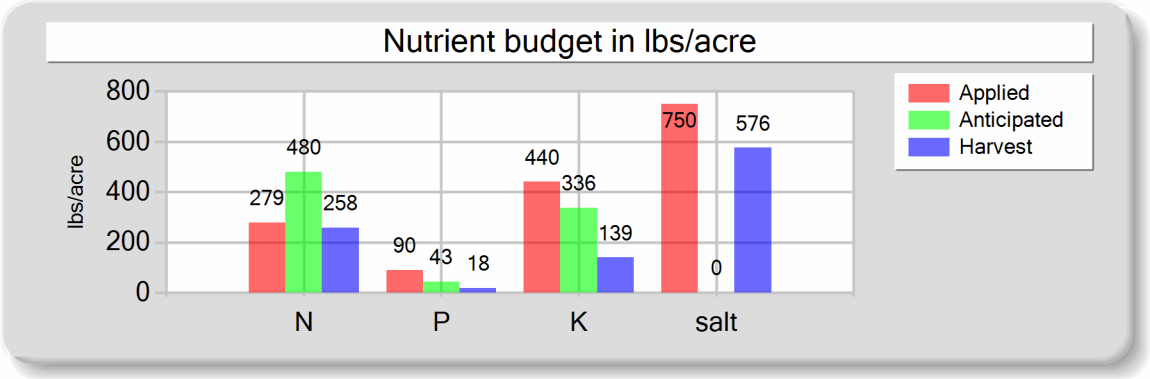
Fresh water applied	
33,270,750.00	*gallons*
1,225.25	*acre-inches*
25.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Little Costa/Schut - 05/27/2020: Alfalfa, hay

Field name: Little Costa/Schut Crop: Alfalfa, hay Plant date: 05/27/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	262.46	90.45	440.39	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.45	0.00	0.00	750.45
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	278.90	90.45	440.39	750.45
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	258.04	18.34	139.14	575.54
Nutrient balance	20.86	72.11	301.25	174.91
Applied to removed ratio	1.08	4.93	3.17	1.30

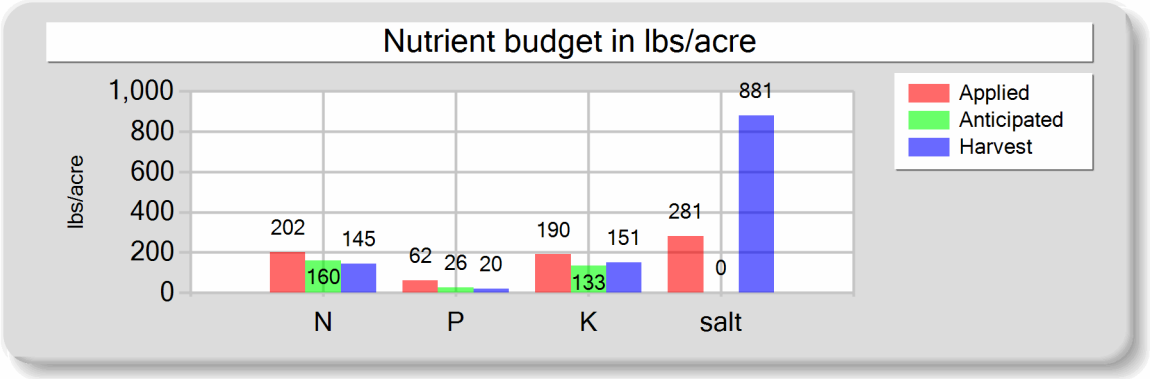
Fresh water applied	
39,098,920.00	*gallons*
1,439.88	*acre-inches*
36.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
6	*harvests*

Poolman - 12/03/2022: Oats, silage-soft dough

Field name: Poolman | Crop: Oats, silage-soft dough | Plant date: 12/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	193.67	62.12	190.02	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.92	0.00	0.00	280.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.59	62.12	190.02	280.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	144.64	19.58	151.17	880.91
Nutrient balance	56.95	42.55	38.85	-599.91
Applied to removed ratio	1.39	3.17	1.26	0.32

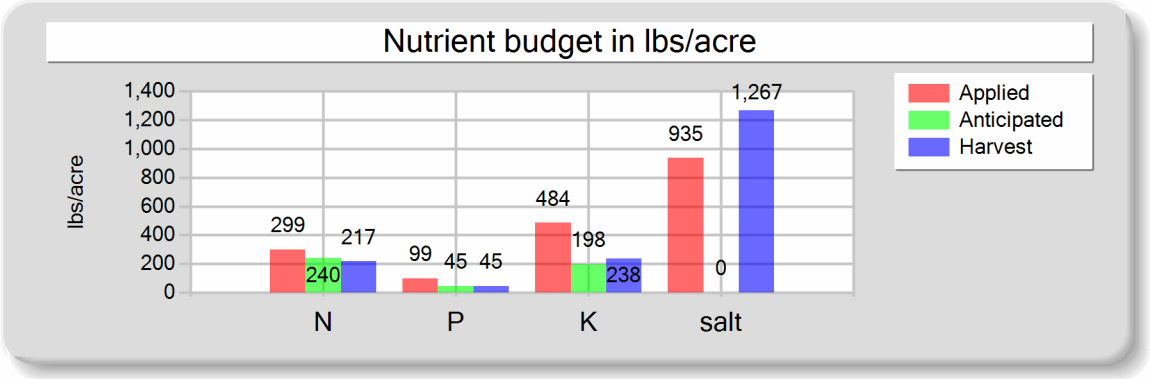
Fresh water applied	
10,980,000.00	*gallons*
404.36	*acre-inches*
13.48	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Poolman - 05/30/2023: Corn, silage

Field name: Poolman Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	288.70	99.50	484.43	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	3.05	0.00	0.00	935.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.75	99.50	484.43	935.33
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.20	45.25	237.56	1,267.00
Nutrient balance	81.55	54.25	246.87	-331.67
Applied to removed ratio	1.38	2.20	2.04	0.74

Fresh water applied	
36,548,600.00	*gallons*
1,345.96	*acre-inches*
44.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid manure

Sample and source description: Solid manure

Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 24.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	5,100.00	15,600.00							0.00
DL	100.00	100.00	100.00							0.01

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 32.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	6,100.00	29,700.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	804.00	204.00	0.00	1.60	85.60	678.00								10,300.00	6,000
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	64.80	39.70	0.00	13.50	41.50	50.20								3,880.00	1,760
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	92.90	72.60	0.00	0.40	22.80	123.00								3,940.00	2,090
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	480.00	407.00	0.00	0.00	60.50	641.00								9,380.00	5,220
DL	1.00	0.50	1.00	1.00	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - 2014 Barn Well

Domestic Well - 2014 Barn Well

2014 Barn Well

Sample description: 2014 Barn Well

Sample date: 11/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	42.00	0.00	42.00								3,240.00	0
DL	0.10	1.00	0.10								10.00	1

Domestic Well - Calf/House Barn

Calf/House Barn

Sample description: Calf/House Barn

Sample date: 11/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	34.90	0.00	34.90								2,890.00	0
DL	1.00	1.00	0.10								10.00	1

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.30	0.00	0.30								133.00	92
DL	0.10	1.00	0.10								10.00	10

Irrigation Water - Dairy Deep Well

Irrigation Water - Dairy Deep Well

Dairy Deep Well

Sample description: Dairy Deep Well

Sample date: 07/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	28.70	0.00	28.70								2,550.00	1,620
DL	0.10	1.00	0.10								10.00	10

Irrigation Water - Lemon Well

Lemon Well

Sample description: Lemon Well

Sample date: 07/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	17.80	0.00	17.10								2,510.00	1,660
DL	1.00	1.00	0.10								10.00	10

Irrigation Water - Magnolia Well

Magnolia Well

Sample description: Magnolia Well

Sample date: 07/26/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20								131.00	96
DL	0.10	1.00	0.10								10.00	10

D. SOIL ANALYSES

Costa Grande

Grande

Sample and source description: Grande

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			129.00				
DL			2.00				

Dairy

Dairy

Sample and source description: Dairy

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			166.00				
DL			2.00				

Greg's

Greg's

Sample and source description: Greg's

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			50.00				
DL			2.00				

Heifer Ranch

Heifer Ranch

Ranch

Sample and source description: Ranch

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			35.00				
DL			2.00				

Little Costa/Schut

Costa/Sch

Sample and source description: Costa/Sch

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			58.00				
DL			2.00				

Poolman

Poolman

Sample and source description: Poolman

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			107.00				
DL			2.00				

Thorkelson

Thorkelson

Thorkelson

Sample and source description: Thorkelson

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			59.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Dairy - 12/03/2022: Oats, silage-soft dough

Dairy

Sample and source description: Dairy

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	2,500.00	15,700.00		10.50
DL	100.00	100.00	100.00		0.01

Dairy - 05/30/2023: Corn, silage

Dairy

Sample and source description: Dairy

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,600.00	2,100.00	11,900.00		5.80
DL	100.00	100.00	100.00		0.01

Greg's - 10/22/2020: Alfalfa, hay

Greg's

Sample and source description: Greg's

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	39,700.00	2,900.00	22,100.00		9.60
DL	10.00	10.00	10.00		0.01

Greg's

Sample and source description: Greg's

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,000.00	2,600.00	9,700.00		6.70
DL	10.00	10.00	10.00		0.01

Heifer Ranch - 12/03/2022: Oats, silage-soft dough

Heifer Ranch A

Sample and source description: Heifer Ranch A

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	2,600.00	16,000.00		11.60
DL	100.00	100.00	100.00		0.01

Heifer Ranch - 12/03/2022: Oats, silage-soft dough

Heifer Ranch B

Sample and source description: Heifer Ranch B

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	1,600.00	13,500.00		8.10
DL	100.00	100.00	100.00		0.01

Heifer Ranch - 05/30/2023: Corn, silage

Heifer Ranch A

Sample and source description: Heifer Ranch A

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,800.00	2,000.00	10,200.00		5.10
DL	100.00	100.00	100.00		0.01

Heifer Ranch B

Sample and source description: Heifer Ranch B

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,500.00	1,700.00	10,200.00		5.30
DL	100.00	100.00	100.00		0.01

Little Costa/Schut - 05/27/2020: Alfalfa, hay

Little Costa/Schut - 05/27/2020: Alfalfa, hay

Little Costa/Schut

Sample and source description: Little Costa/Schut

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	40,800.00	2,900.00	22,000.00		9.10
DL	100.00	100.00	100.00		0.01

Little Costa/Schut

Sample and source description: Little Costa/Schut

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,100.00	2,200.00	15,900.00		7.00
DL	100.00	100.00	100.00		0.01

Poolman - 12/03/2022: Oats, silage-soft dough

Poolman

Sample and source description: Poolman

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	1,800.00	13,900.00		8.10
DL	100.00	100.00	100.00		0.01

Poolman - 05/30/2023: Corn, silage

Poolman - 05/30/2023: Corn, silage

Poolman

Sample and source description: Poolman

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,600.00	2,000.00	10,500.00		5.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

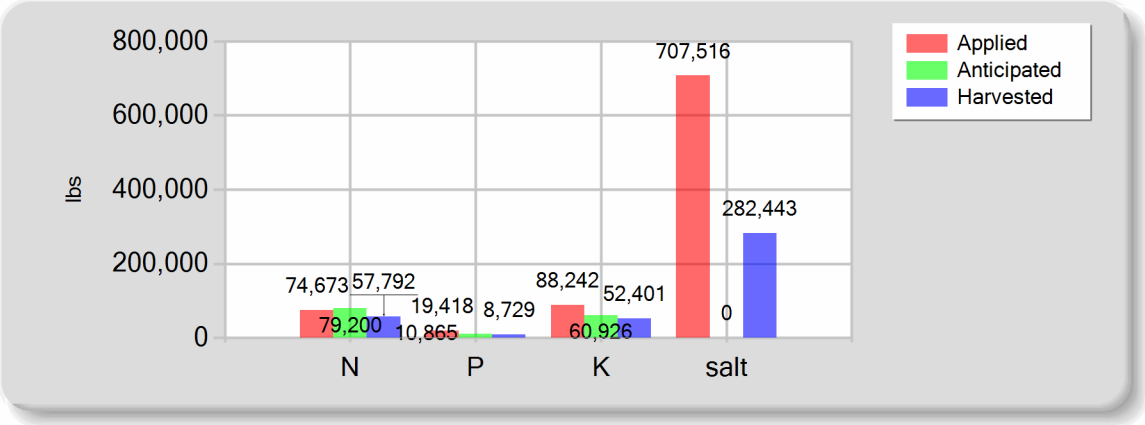
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

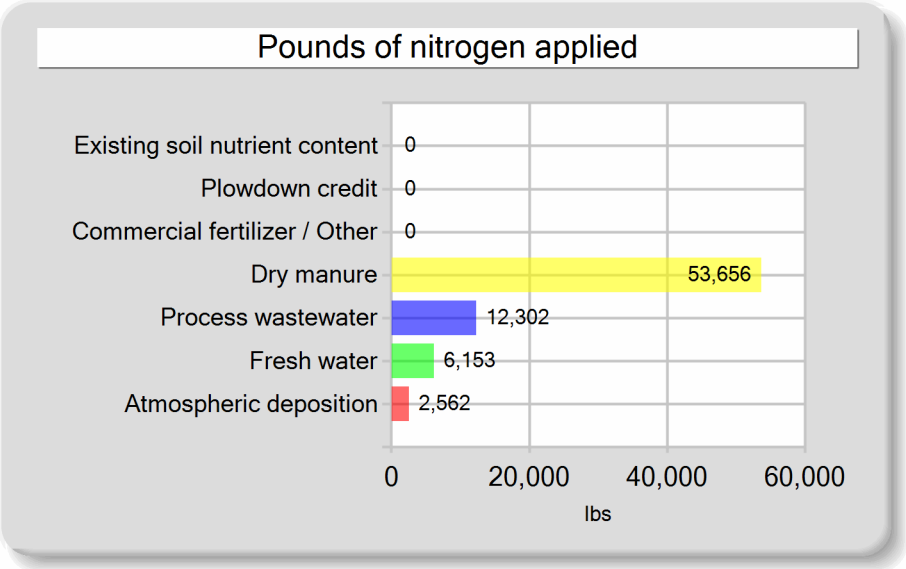
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

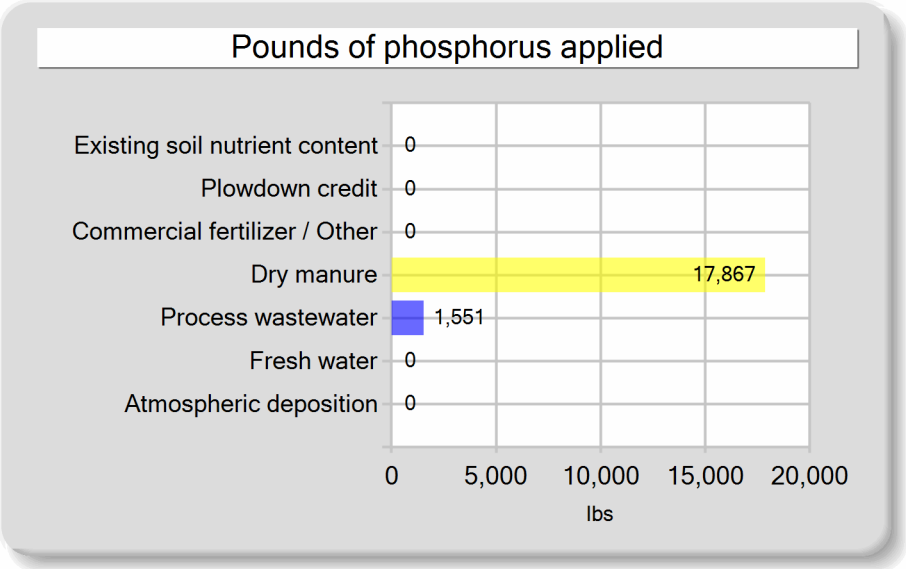
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	53,655.92	17,867.28	71,813.53	0.00
Process wastewater	12,302.20	1,550.59	16,428.56	133,786.41
Fresh water	6,152.66	0.00	0.00	573,729.79
Atmospheric deposition	2,562.00	0.00	0.00	0.00
Total nutrients applied	74,672.78	19,417.87	88,242.09	707,516.20
Anticipated crop nutrient removal	79,200.00	10,864.80	60,926.40	0.00
Actual crop nutrient removal	57,792.41	8,729.46	52,401.31	282,442.66
Nutrient balance	16,880.36	10,688.41	35,840.78	425,073.54
Applied to removed ratio	1.29	2.22	1.68	2.50

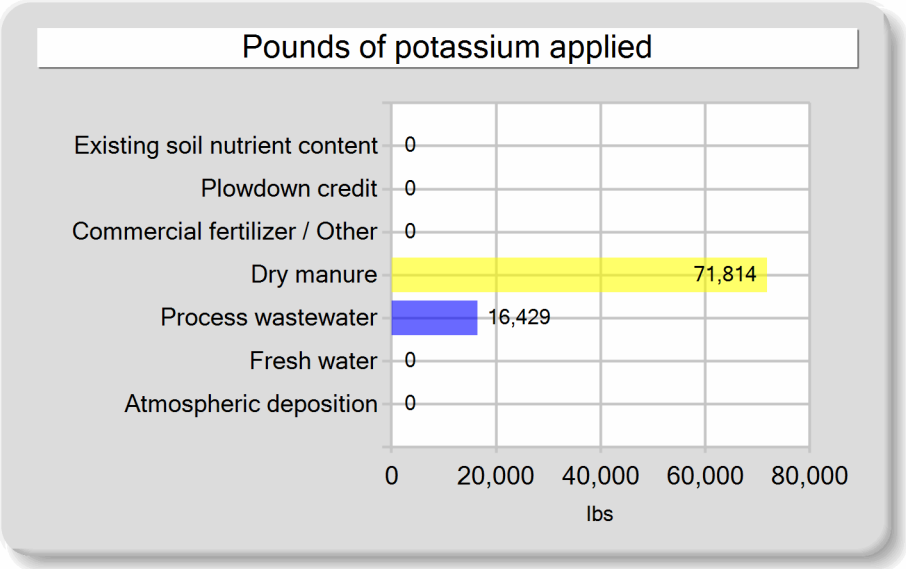
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

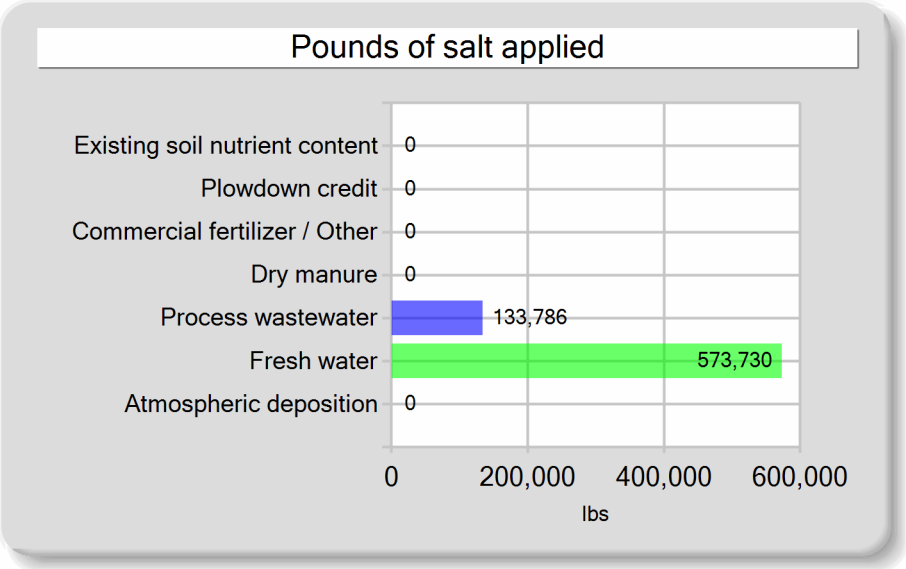


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Azevedo
John Azevedo (Jun 21, 2024 15:55 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John M Azevedo	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/21/2024	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **John Azevedo**

Name of Dairy Facility: **Azevedo Dairy, Inc.**

Facility Address: **1337 Magnolia Avenue, Patterson, CA 95363**

Contact Person Name and Phone Number: **John Azevedo (209) 892-6961**

Manure Hauler Information:

Name of Hauling Company/Person: **Azevedo Dairy, Inc.**

Address of Hauling Company /Person: **1337 Magnolia Avenue, Patterson CA 95363**

Contact Person and Phone Number: **John Azevedo (209) 892-6961**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
JCA Custom Farming	1343 Magnolia Avenue	Patterson	95363	(209) 495-3055

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
1343 Magnolia Avenue	Patterson	95363	

Dates Hauled: **5/16/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **1,800** Tons

Manure Solids Content (if amount reported in tons): **75.5%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: John Azevedo
John Azevedo (Jun 21, 2024 15:55 PDT) Date: 06/21/2024

Hauler's Signature: John Azevedo
John Azevedo (Jun 21, 2024 15:55 PDT) Date: 06/21/2024

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **John Azevedo**

Name of Dairy Facility: **Azevedo Dairy, Inc.**

Facility Address: **1337 Magnolia Avenue, Patterson, CA 95363**

Contact Person Name and Phone Number: **John Azevedo (209) 892-6961**

Manure Hauler Information:

Name of Hauling Company/Person: **Azevedo Dairy, Inc.**

Address of Hauling Company /Person: **1337 Magnolia Avenue, Patterson CA 95363**

Contact Person and Phone Number: **John Azevedo (209) 892-6961**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
JCA Custom Farming	1343 Magnolia Avenue	Patterson	95363	(209) 495-3055

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
1343 Magnolia Avenue	Patterson	95363	

Dates Hauled: **10/8/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **2,000** Tons

Manure Solids Content (if amount reported in tons): **67.4%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: John Azevedo
John Azevedo (Jun 21, 2024 15:55 PDT) Date: 06/21/2024

Hauler's Signature: John Azevedo
John Azevedo (Jun 21, 2024 15:55 PDT) Date: 06/21/2024

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Azevedo Dairy, Inc.

Received: 07/27/2023 16:00
Reported: 08/04/2023 15:12

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23G1990-01	Canal	Ag Water	T. Madruga		07/26/2023 14:30
23G1990-02	Dairy Deep Well	Ag Water	T. Madruga		07/26/2023 15:00
23G1990-03	Lemon Well	Ag Water	T. Madruga		07/26/2023 15:30
23G1990-04	Magnolia Well	Ag Water	T. Madruga		07/26/2023 16:00

Default Cooler Temperature on Receipt °C: 1.7
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Reported: 08/04/2023 15:12

Sample Results

Sample: Canal
23G1990-01 (Water)

Sampled: 7/26/2023 14:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.13**	mmhos/cm	0.01	1		07/31/23 13:11	SM 2510 B		BEG1120
Electrical Conductivity umhos	**133**	umhos/cm	10.0	1		07/31/23 13:11	SM 2510 B		BEG1120
Nitrate Nitrogen as NO3N	**0.3**	mg/L	0.1	1	10	07/27/23 22:47	EPA 300.0		BEG1056
pH	**7.5**	units	1.0	1		07/31/23 13:11	SM 4500-H+	H	BEG1120
Total Filterable Solids (TDS)	**92.0**	mg/L	10.0	1		08/03/23 13:15	SM 2540 C		BEG1153
Temperature	**25.0**	°C	0.0	1		07/31/23 13:11	SM 2510 B		BEG1120
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/03/23 08:58	SM 4500-NH3 C		BEH0061
Total Nitrogen	ND	mg/L	1.00	1		08/03/23 08:58	SM 4500-NH3 C		BEH0061

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Received: 07/27/2023 16:00
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Sample Results
(Continued)

Sample: Dairy Deep Well
23G1990-02 (Water)

Sampled: 7/26/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.55**	mmhos/cm	0.01	1		07/31/23 13:12	SM 2510 B		BEG1120
Electrical Conductivity umhos	**2550**	umhos/cm	10.0	1		07/31/23 13:12	SM 2510 B		BEG1120
Nitrate Nitrogen as NO3N	**28.7**	mg/L	0.1	1	10	07/27/23 23:08	EPA 300.0		BEG1056
pH	**7.6**	units	1.0	1		07/31/23 13:12	SM 4500-H+	H	BEG1120
Total Filterable Solids (TDS)	**1620**	mg/L	10.0	1		08/03/23 13:15	SM 2540 C		BEG1153
Temperature	**25.0**	°C	0.0	1		07/31/23 13:12	SM 2510 B		BEG1120
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/03/23 08:59	SM 4500-NH3 C		BEH0061
Total Nitrogen	**29.2**	mg/L	1.00	1		08/03/23 08:59	SM 4500-NH3 C		BEH0061

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Sample Results
(Continued)

Sample: Lemon Well
23G1990-03 (Water)

Sampled: 7/26/2023 15:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.51**	mmhos/cm	0.01	1		07/31/23 13:13	SM 2510 B		BEG1120
Electrical Conductivity umhos	**2510**	umhos/cm	10.0	1		07/31/23 13:13	SM 2510 B		BEG1120
Nitrate Nitrogen as NO3N	**17.1**	mg/L	0.1	1	10	07/27/23 23:29	EPA 300.0		BEG1056
pH	**7.7**	units	1.0	1		07/31/23 13:13	SM 4500-H+	H	BEG1120
Total Filterable Solids (TDS)	**1660**	mg/L	10.0	1		08/03/23 13:15	SM 2540 C		BEG1153
Temperature	**25.0**	°C	0.0	1		07/31/23 13:13	SM 2510 B		BEG1120
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/03/23 09:47	SM 4500-NH3 C		BEH0086
Total Nitrogen	**17.8**	mg/L	1.00	1		08/03/23 09:47	SM 4500-NH3 C		BEH0086

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Sample Results
(Continued)

Sample: Magnolia Well
23G1990-04 (Water)

Sampled: 7/26/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.13**	mmhos/cm	0.01	1		07/31/23 13:15	SM 2510 B		BEG1120
Electrical Conductivity umhos	**131**	umhos/cm	10.0	1		07/31/23 13:15	SM 2510 B		BEG1120
Nitrate Nitrogen as NO3N	**0.2**	mg/L	0.1	1	10	07/27/23 23:50	EPA 300.0		BEG1056
pH	**8.0**	units	1.0	1		07/31/23 13:15	SM 4500-H+	H	BEG1120
Total Filterable Solids (TDS)	**96.0**	mg/L	10.0	1		08/03/23 13:15	SM 2540 C		BEG1153
Temperature	**25.0**	°C	0.0	1		07/31/23 13:15	SM 2510 B		BEG1120
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/03/23 09:49	SM 4500-NH3 C		BEH0086
Total Nitrogen	ND	mg/L	1.00	1		08/03/23 09:49	SM 4500-NH3 C		BEH0086

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1056										
Blank (BEG1056-BLK1)					Prepared & Analyzed: 7/27/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEG1056-BLK2)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEG1056-BS1)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.8	90-110		
Duplicate (BEG1056-DUP1)		**Source: 23G2094-01**			Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L		5.0			0.343	10
Matrix Spike (BEG1056-MS1)		**Source: 23G2094-01**			Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	10.5		0.1	mg/L	5.000	5.0	111	90-110		
Reference (BEG1056-SRM1)					Prepared & Analyzed: 7/27/2023					
Nitrate Nitrogen as NO3N	9.5			mg/L	10.00		95.4	90-110		
Reference (BEG1056-SRM2)					Prepared: 7/27/2023 Analyzed: 7/28/2023					
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		96.2	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1120										
Blank (BEG1120-BLK1)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	5.4		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEG1120-BLK2)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	6.0		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEG1120-BLK3)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.9		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEG1120-DUP1)		**Source: 23G1984-01**			Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	7.3		1.0	units		7.3			0.274	10
Electrical Conductivity	1.14		0.01	mmhos/cm		1.15			0.750	10
Electrical Conductivity umhos	1140		10.0	umhos/cm		1150			0.750	10
Duplicate (BEG1120-DUP2)		**Source: 23G1988-02**			Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	1.05		0.01	mmhos/cm		1.05			0.0475	10
pH	7.6		1.0	units		7.6			0.131	10
Electrical Conductivity umhos	1050		10.0	umhos/cm		1050			0.0475	10
Reference (BEG1120-SRM1)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	527			umhos/cm	538.0		98.0	90-110		
Reference (BEG1120-SRM2)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	7.8			units	7.790		100	.7163-101.28		
Reference (BEG1120-SRM3)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	981			umhos/cm	1000		98.1	90-110		
Electrical Conductivity umhos	981			umhos/cm	1000		98.1	90-110		
Reference (BEG1120-SRM4)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	979			umhos/cm	1000		97.9	90-110		
Electrical Conductivity umhos	979			umhos/cm	1000		97.9	90-110		
Reference (BEG1120-SRM5)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity	979			umhos/cm	1000		97.9	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1120 (Continued)										
Reference (BEG1120-SRM5)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
Electrical Conductivity umhos	979			umhos/cm	1000		97.9	90-110		
Reference (BEG1120-SRM6)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEG1120-SRM7)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEG1120-SRM8)					Prepared: 7/28/2023	Analyzed: 7/31/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEG1153										
Blank (BEG1153-BLK1)					Prepared: 7/31/2023	Analyzed: 8/3/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEG1153-BS1)					Prepared: 7/31/2023	Analyzed: 8/3/2023				
Total Filterable Solids (TDS)	15.0		10.0	mg/L	2000		0.750	0-200		
Duplicate (BEG1153-DUP1)		**Source: 23G2009-01**			Prepared: 7/31/2023	Analyzed: 8/3/2023				
Total Filterable Solids (TDS)	3800		10.0	mg/L		3820			0.656	10
Reference (BEG1153-SRM1)					Prepared: 7/31/2023	Analyzed: 8/3/2023				
Total Filterable Solids (TDS)	310			mg/L	325.0		95.4	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0061										
Blank (BEH0061-BLK1)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH0061-BLK2)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH0061-BS1)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	5.46		1.00	mg/L	5.709		95.6	90-110		
LCS (BEH0061-BS2)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	5.65		1.00	mg/L	5.709		99.0	90-110		
Duplicate (BEH0061-DUP1)		**Source: 23G1990-02**			Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BEH0061-DUP2)		**Source: 23H0322-04**			Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH0061-MS1)		**Source: 23G1990-02**			Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	7.94		1.40	mg/L	7.992	ND	99.4	90-110		
Matrix Spike (BEH0061-MS2)		**Source: 23H0322-04**			Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	7.46		1.40	mg/L	7.992	ND	93.3	90-110		
Reference (BEH0061-SRM1)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	23.6			mg/L	23.80		99.3	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0086										
Blank (BEH0086-BLK1)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH0086-BLK2)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH0086-BS1)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	5.58		1.00	mg/L	5.709		97.8	90-110		
LCS (BEH0086-BS2)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	5.15		1.00	mg/L	5.709		90.2	90-110		
Duplicate (BEH0086-DUP1)		**Source: 23G1867-02**			Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	96.1		3.50	mg/L		92.3			4.03	10
Duplicate (BEH0086-DUP2)		**Source: 23G2034-01**			Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	115		3.50	mg/L		114			0.196	10
Matrix Spike (BEH0086-MS1)		**Source: 23G1867-02**			Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	106		3.50	mg/L	9.990	92.3	139	90-110		
Matrix Spike (BEH0086-MS2)		**Source: 23G2034-01**			Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	119		3.50	mg/L	9.990	114	41.2	90-110		
Reference (BEH0086-SRM1)					Prepared: 8/2/2023	Analyzed: 8/3/2023				
Kjeldahl Nitrogen (TKN), Total	23.6			mg/L	23.80		99.0	90-110		

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07/27/23 16:00 23G1990

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 4 No. of Bottles: 8

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Azevedo Dairy, Inc.
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

( ) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
(X) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Canal	7/26/2023	1430	1.7	
2	Dairy Deep Well	7/26/2023	1500	2.6	
3	Lemon Well	7/26/2023	1530	3.1	
4	Magnolia Well	7/26/2023	1600	2.5	
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	7/26/2023 @ 1600	7/27/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	7/27/2023 @ 900	7-27-23 3:35
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

07/27/23 16:00 23G1990

Shipping Information: Fed Ex □ UPS □ GSO □ Picked-Up ☑ Walk In □ DLI Sampler □
Ontrac □ PMS □ USPS □ Other □ _____

Shipping Container: Ice Chest ☑ None □ Box □

Refrigerant: Wet Ice: ☑ Blue Ice: □ None: □

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)	1<2	1<2	1<2	1<2						
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1	1						
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Azevedo Dairy, Inc.

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:43

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0482-01	2014 Barn Well	Ag Water	T. Madruga		11/08/2023 13:00
23K0482-02	Calf/House Barn	Ag Water	T. Madruga		11/08/2023 13:30

Default Cooler Temperature on Receipt °C: -0.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Azevedo Dairy, Inc.

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:43

Sample Results

Sample: 2014 Barn Well
23K0482-01 (Water)

Sampled: 11/8/2023 13:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**3.24**	mmhos/cm	0.01	1		11/15/23 11:16	SM 2510 B		BEK0350
Electrical Conductivity umhos	**3240**	umhos/cm	10.0	1		11/15/23 11:16	SM 2510 B		BEK0350
Ammonia (as N)	ND	mg/L	0.00	1		11/08/23 13:00	Field		BEK0301
Nitrate Nitrogen as NO3N	**42.0**	mg/L	0.1	1	10	11/09/23 20:22	EPA 300.0		BEK0247
Temperature	**25.0**	units	0.0	1		11/15/23 11:16	SM 4500-H+	H	BEK0350
pH	**7.5**	units	1.0	1		11/15/23 11:16	SM 4500-H+	H	BEK0350

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Azevedo Dairy, Inc.

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:43

Sample Results
(Continued)

Sample: Calf/House Barn
23K0482-02 (Water)

Sampled: 11/8/2023 13:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.89**	mmhos/cm	0.01	1		11/15/23 11:17	SM 2510 B		BEK0350
Electrical Conductivity umhos	**2890**	umhos/cm	10.0	1		11/15/23 11:17	SM 2510 B		BEK0350
Ammonia (as N)	ND	mg/L	0.00	1		11/08/23 13:30	Field		BEK0301
Nitrate Nitrogen as NO3N	**34.9**	mg/L	0.1	1	10	11/09/23 23:02	EPA 300.0		BEK0247
Temperature	**25.0**	units	0.0	1		11/15/23 11:17	SM 4500-H+	H	BEK0350
pH	**7.7**	units	1.0	1		11/15/23 11:17	SM 4500-H+	H	BEK0350

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Azevedo Dairy, Inc.

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:43

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0247										
Blank (BEK0247-BLK1)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0247-BLK2)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0247-BLK3)					Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0247-BS1)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		101	90-110		
LCS (BEK0247-BS2)					Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.5	90-110		
Duplicate (BEK0247-DUP1)		**Source: 23K0409-01**			Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L		ND				10
Duplicate (BEK0247-DUP2)		**Source: 23K0487-01**			Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	2.1		0.1	mg/L		2.1			1.25	10
Matrix Spike (BEK0247-MS1)		**Source: 23K0409-01**			Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000	ND	97.5	90-110		
Matrix Spike (BEK0247-MS2)		**Source: 23K0487-01**			Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	6.9		0.1	mg/L	5.000	2.1	97.2	90-110		
Reference (BEK0247-SRM1)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.0	90-110		
Reference (BEK0247-SRM2)					Prepared & Analyzed: 11/9/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.6	90-110		
Reference (BEK0247-SRM3)					Prepared: 11/9/2023 Analyzed: 11/10/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.1	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Azevedo Dairy, Inc.

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:43

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0350										
Blank (BEK0350-BLK1)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.6		1.0	units						
Blank (BEK0350-BLK2)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEK0350-BLK3)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Duplicate (BEK0350-DUP1)		**Source: 23K0483-01**			Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	3.23		0.01	mmhos/cm		3.27			1.28	10
pH	7.4		1.0	units		7.4			0.269	10
Electrical Conductivity umhos	3230		10.0	umhos/cm		3270			1.28	10
Duplicate (BEK0350-DUP2)		**Source: 23K0518-04**			Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	0.68		0.01	mmhos/cm		0.68			0.337	10
Electrical Conductivity umhos	685		10.0	umhos/cm		682			0.337	10
pH	7.8		1.0	units		7.8			0.256	10
Reference (BEK0350-SRM1)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	576			umhos/cm	538.0		107	90-110		
Reference (BEK0350-SRM2)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0350-SRM3)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0350-SRM4)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0350-SRM5)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Azevedo Dairy, Inc.

Received: 11/09/2023 16:20
Reported: 11/16/2023 10:43

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0350 (Continued)										
Reference (BEK0350-SRM5)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEK0350-SRM6)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0350-SRM7)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0350-SRM8)					Prepared: 11/10/2023	Analyzed: 11/15/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/09/23 16:20 23K0482

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 2 No. of Bottles: 2

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State: CA** **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Azevedo Dairy, Inc.
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	2014 Barn Well	11/8/2023	1300	-0.4	ND
2	Calf/House Barn	11/8/2023	1330	-0.1	ND
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/8/2023 @ 1330	11/9/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/9/2023 @9am	11-9-23 3:55
Third	[illegible]	[illegible]	11/9 16:20	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
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Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/09/23 16:20 23K0482

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice: ☑ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										